BEGINNER RUN GUIDE

Suicide-Risk Detection Using NLP-Based Text and Speech Analysis

> **Final status:** Implementation-ready research prototype COMPLETE. Fresh final verification: **37/37 tests PASS, 0 failed**. Text-only, Speech-only, and Text + Speech outputs were all verified.

This document is written so that even a person with no previous knowledge of Python, Streamlit, machine learning, or this project can install the software, verify the files, run the project, test all three modes, understand the output, and solve common errors.

1. Verified Project Location

Main project:

```
C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000
```

Backup:

```
C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000_BACKUP_2026-10-29
```

If the project is moved to another computer, copy the **entire Thanmayi 8000 folder**. Do not move only the model files or only the source-code folder.

2. What the Project Does

Text Only

Text is converted using TF-IDF unigram/bigram representation and classified with the current best verified text model: **Calibrated Linear SVM**. Output: `P_text`.

Speech Only

Speech is loaded as mono 16 kHz audio, peak-normalized, and converted into acoustic features:

- MFCC
- pitch
- RMS energy
- zero-crossing rate
- spectral centroid
- duration

The current operational speech model is a **Logistic Regression** model selected using participant-grouped cross-validation. Output: `P_speech`.

Text + Speech

The application combines branch scores using a transparent prototype late-fusion rule:

- Text weight = 0.70
- Speech weight = 0.30
- Low: score < 0.40
- Moderate: 0.40 to < 0.70
- High: >= 0.70

> **Important:** This is a research screening prototype, not a clinical diagnosis system.

3. Computer Requirements

Recommended:

- Windows 10 or Windows 11, 64-bit
- 8 GB RAM minimum; 16 GB recommended
- 10 GB free disk space
- Chrome or Microsoft Edge
- Internet connection for first-time package installation

After the packages are installed, normal local running does not require continuous internet access.

4. Software to Install

4.1 Python — REQUIRED

Recommended version: **Python 3.11.x**

Official download: https://www.python.org/downloads/

Windows releases: https://www.python.org/downloads/windows/

During installation:

1. Enable **Add Python to PATH** if shown.
2. Finish installation.
3. Close and reopen Command Prompt.

Verify with one of these:

```
python --version
```

```
python3 --version
```

```
py -3.11 --version
```

4.2 Browser — REQUIRED

Use Google Chrome or Microsoft Edge.

4.3 Visual Studio Code — OPTIONAL

Only needed if source code must be viewed or edited.

Official: https://code.visualstudio.com/

Windows guide: https://code.visualstudio.com/docs/setup/windows

4.4 Git — OPTIONAL

Not required when the complete project folder is already available.

Official: https://git-scm.com/install/windows

4.5 FFmpeg — OPTIONAL

For the safest demo, use WAV audio. The verified demo works with WAV.

FFmpeg is only useful if another audio format cannot be decoded on a particular computer.

Official: https://ffmpeg.org/

5. Python Packages

The project already contains `requirements.txt` with:

- pandas 2.3.3
- numpy 2.4.4
- scikit-learn 1.8.0
- joblib 1.5.3
- librosa 0.11.0
- soundfile 0.14.0
- streamlit 1.61.0
- torch 2.11.0
- transformers 5.5.0
- matplotlib 3.11.2
- flask 3.1.3

Do not install these manually one-by-one unless troubleshooting.

6. First-Time Setup — Easiest Method

Open:

`C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000`

Double-click:

`SETUP_PROJECT.bat`

The script automatically:

1. finds Python
2. upgrades pip
3. installs `requirements.txt`
4. shows **SETUP COMPLETED SUCCESSFULLY** when finished

This is normally required only once on a new PC.

7. Manual Setup Method

Open Command Prompt.

Run:

```
cd /d "C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000"
```

Then:

```
python -m pip install --upgrade pip
```

```
python -m pip install -r requirements.txt
```

If `python` does not work, use:

```
python3 -m pip install -r requirements.txt
```

or:

```
py -3.11 -m pip install -r requirements.txt
```

8. Verify the Project

Double-click:

```
VERIFY_PROJECT.bat
```

This runs the automated project checks.

Expected verified final result:

- Passed: **37**
- Failed: **0**
- Overall: **PASS**

The verification checks:

- required files
- Python syntax
- dataset counts
- valid labels
- train-test leakage = 0
- stored model metrics
- speech metrics
- 6 diagrams
- text inference
- real EATD WAV speech inference
- fusion output

9. Start the Project — Easiest Method

Double-click:

`START_PROJECT.bat`

Keep the terminal open.

The browser will open the Streamlit application.

If it does not open automatically, open:

http://localhost:8501

10. Start Manually

Open Command Prompt:

```
cd /d "C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000"
```

Then run:

```
python -m streamlit run app.py
```

Alternative:

```
python3 -m streamlit run app.py
```

or:

```
py -3.11 -m streamlit run app.py
```

11. Text-Only Mode

1. Select **Text only**
2. Enter text
3. Click **Analyze**
4. Read:
    - Screening Level
    - Combined Evidence Score
    - `P_text`

Verified example:

```
I am feeling very stressed and overwhelmed today.
```

Actual verified output:

- `P_text = 0.194`
- Screening Level = **Low**

12. Speech-Only Mode

1. Select **Speech only**
2. Upload a WAV
3. Click **Analyze**
4. Read:
    - Screening Level

- `P_speech`

Verified test audio:

`C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000\Datasets\validation\10\negative_out.wav`

Actual verified output:

- `P_speech = 0.282`
- Screening Level = **Low**

13. Text + Speech Mode

1. Select **Text + Speech**
2. Enter text
3. Upload WAV
4. Click **Analyze**

Verified output:

- `P_text = 0.194`
- `P_speech = 0.282`
- Combined score = `0.221`
- Screening Level = **Low**

Formula:

`Combined = 0.70 × P_text + 0.30 × P_speech`

14. Verified Output Screenshots

Text Only



Speech Only



Text + Speech



15. Stop the Application

Go to the terminal where Streamlit is running.

Press:

```
Ctrl + C
```

Then close the terminal.

16. Important Project Files

- `app.py` — Streamlit UI
- `src\inference_pipeline.py` — text/speech inference
- `models\text_model\` — text model artifacts
- `models\speech_model\` — speech model artifacts
- `config\fusion_config.json` — late-fusion configuration
- `Datasets\` — datasets and processed files
- `reports\` — metrics, screenshots, final test report
- `diagrams\` — six system-design diagrams
- `tests\run_project_tests.py` — automated verification
- `SETUP_PROJECT.bat` — first-time installation helper
- `START_PROJECT.bat` — app launcher
- `VERIFY_PROJECT.bat` — verification helper

17. Verified Results

Text Model — Current Best

TF-IDF + Calibrated Linear SVM:

- Accuracy 72.56%
- Precision 71.92%
- Recall 75.65%
- F1 73.74%
- ROC-AUC 78.22%

Speech Model

Official EATD participant-aggregated validation:

- Accuracy 63.29%
- Balanced Accuracy 63.44%
- Precision 21.88%
- Recall 63.64%
- F1 32.56%
- ROC-AUC 59.22%

18. Dataset Facts

SDCNL

Training:

- 1,513 rows
- Class 0 = 727
- Class 1 = 786

Testing:

- 379 rows
- Class 0 = 186
- Class 1 = 193

Normalized train-test overlap:

- **0**

EATD

- Train folders = 83
- Usable train participants = 80
- Validation folders = 79
- Usable validation participants = 79
- Train audio feature rows = 240
- Validation audio feature rows = 237

19. Common Errors

Python is not recognized

Install Python 3.11 and make sure Python is added to PATH.

No module named streamlit

Run:

```
python -m pip install -r requirements.txt
```

Port 8501 already in use

Close old Streamlit windows or run:

```
python -m streamlit run app.py --server.port 8502
```

Then open:

http://localhost:8502

Audio decoding error

Use WAV first. Verified example: `Datasets\validation\10\negative_out.wav`

MemoryError

Close unnecessary apps and old Python processes. For normal demonstration, do not run training scripts.

Model file not found

Make sure the **entire project folder** was copied. Do not move `models` separately.

20. What Is NOT Required for a Normal Demo

Do not retrain:

- BiLSTM
- BERT embeddings
- SVM
- speech model

Do not:

- merge SDCNL and EATD row-by-row
- rerun preprocessing
- create a database
- edit model files

The trained model artifacts are already saved.

21. Viva Explanation

Always explain:

This is a research screening prototype and not a clinical diagnostic system.

Reason:

- SDCNL and EATD are not participant-matched
- EATD is depression/distress data, not direct suicide labels
- fusion weights are prototype settings
- therefore no unsupported final multimodal accuracy is claimed

22. Quick Demo Checklist

1. Run `VERIFY_PROJECT.bat`
2. Confirm PASS
3. Run `START_PROJECT.bat`
4. Demonstrate Text Only
5. Demonstrate Speech Only
6. Demonstrate Text + Speech
7. Explain `P_text`
8. Explain `P_speech`
9. Explain late fusion
10. Show diagrams if asked
11. Stop with `Ctrl + C`

23. Final Completion Checklist

- Dataset audit: COMPLETE
- Text preprocessing: COMPLETE
- Leakage verification: COMPLETE
- Text models: COMPLETE
- Speech preprocessing: COMPLETE
- Speech model: COMPLETE
- Text inference: COMPLETE
- Speech inference: COMPLETE

- Late fusion: COMPLETE as prototype
- Streamlit UI: COMPLETE
- Six diagrams: COMPLETE
- Diagram ZIP: COMPLETE
- Automated testing: COMPLETE
- Final tests: **37/37 PASS**
- Three output screenshots: COMPLETE
- Final report: COMPLETE
- Beginner run guide: COMPLETE

> **FINAL PROJECT IMPLEMENTATION STATUS: COMPLETE AS A RESEARCH PROTOTYPE.**